One application through GitHub Actions

DevOps & infrastructure | Intermediate

Session M4-01 | 3 hours

Agenda

Time	Topic
0:00 to 0:08	Baseline and Actions model
0:08 to 0:18	Generate CI
0:18 to 0:28	Design deployment
0:28 to 0:40	Repair workflow failures
0:40 to 0:52	Validate a JavaScript action
0:52 to 1:00	Review and lab handoff

Start with repository facts

`copilot-webapp` already defines the contract:

```
npm ci
npm test
npm run lint
npm run build
test -f dist/app.js
```

The workflow should call these commands. It should not invent another build, artifact path, or runtime.

A workflow is code with an execution boundary

Review	Question
Trigger	Who can cause this code to run?
Runner	Which tools and trust boundary apply?
Permissions	What can the token read or change?
Commands	Do they exist in the repository?
Artifacts	Which files leave the job?
Environment	Which approval gate applies?

Generate CI from the application

```
Read package.json, tests, and scripts/build.js. Create ci.yml for pull requests
and pushes to main. Use Node.js 20, npm ci, existing lint, test, and build scripts,
npm caching, contents: read, and a seven-day dist artifact.
```

Then check both YAML syntax and application behavior:

```
ruby -e 'require "yaml"; Psych.parse_stream(File.read(ARGV[0]))' ci.yml
npm test && npm run lint && npm run build
```

Deployment stays a design exercise

The same application moves through:

```
build → staging → production
```

The workflow references environments and secrets. Production reviewers are configured in repository settings. Placeholder commands must not imply that a deployment ran.

Syntax and behavior fail differently

`broken-ci.yml` does not parse.

`broken-deploy.yml` parses but contains a stale runner, a bad Git ref, a wrong secret name, and no checkout.

Ask for the failed-step output before the repair.

A repair is a claim plus proof

```
Observation: the parser fails near Setup Node.js.
Change: align the step indentation.
Proof: parse the file again.
```

For behavior faults, cite the deployment specification or repository facts.

Validate both action outcomes

```
env INPUT_THRESHOLD=0 ... node src/main.js
env INPUT_THRESHOLD=101 ... node src/main.js
```

The first run must pass and write a summary. The second must exit with code 1.

One successful run does not show that the action enforces the threshold.

Access and checks

GitHub Copilot access is required. Repository access and Actions are optional.

For each case, inspect:

- command or review;
- result;
- changed files;
- the next action.

Lab

Create CI, design deployment, repair failures, and validate the action against `copilot-webapp`.